

# San José State University

## College of Engineering — Graduate and Extended Studies

### CMPE 272, Enterprise Software Platforms — Sections 03 & 49 — Fall 2026

#### Course and Contact Information

|                                 |                                                            |
|---------------------------------|------------------------------------------------------------|
| <b>Instructor:</b>              | Andrew Bond                                                |
| <b>Office Location:</b>         | E281, or via Zoom                                          |
| <b>Telephone:</b>               | 408-666-7339                                               |
| <b>Email:</b>                   | andrew.bond@sjsu.edu                                       |
| <b>Office Hours:</b>            | Tuesdays, 4:30 - 6:00 PM, or by appointment                |
| <b>Section 03</b> (Class 47075) | <b>Tuesdays, 6:00 – 8:45 PM</b> — Engineering Building 333 |
| <b>Section 49</b> (Class 47716) | <b>Wednesdays, 6:00 – 8:45 PM</b> — Hybrid                 |
| <b>Class Dates:</b>             | Aug 19 – Dec 7, 2026                                       |
| <b>Prerequisites:</b>           | Classified graduate standing or graduate advisor consent   |
| <b>Canvas Website:</b>          | FA26: CMPE 272-03 and CMPE 272-49                          |

Both sections cover the same material and are assessed identically. Section 49 is hybrid; the in-person/online pattern is posted to Canvas.

---

## Course Description

Enterprise software platforms: the technologies, architectures and practices used to build and operate software at organizational scale. The course covers the enterprise stack from infrastructure and application frameworks through data platforms, enterprise applications, AI integration, security and operations, with a substantial team project.

This offering rebalances the sequence toward where enterprise engineering effort now concentrates — platform engineering, data engineering, and the integration of AI systems into existing enterprise estates — while keeping the architectural fundamentals that make those choices legible.

---

## Course Learning Objectives (CLO)

Upon successful completion of this course, students will be able to:

- **CLO1:** Describe the enterprise software stack end to end and explain how its layers interact.
- **CLO2:** Evaluate application frameworks, APIs and integration patterns against a stated enterprise requirement.
- **CLO3:** Compare cloud-native deployment options — containers, serverless, service mesh — and justify a choice on cost, complexity and operational burden.
- **CLO4:** Explain the role of ERP, SCM and CRM systems in an enterprise and how they are extended and integrated.
- **CLO5:** Model a business process and compose services to implement it.
- **CLO6:** Select an appropriate datastore for an access pattern and design a data pipeline over it.
- **CLO7:** Integrate AI and machine-learning capability into an enterprise application, including retrieval over enterprise data, and describe its evaluation and governance.
- **CLO8:** Assess enterprise security posture across identity, zero trust and software supply-chain integrity.
- **CLO9:** Instrument a system for resilience and observability, and reason about SLOs and incident response.
- **CLO10:** Deliver a working team project with a live demonstration, a code submission and a written report.

---

## Required Texts / Readings

Readings are drawn from O'Reilly Safari Online (available through the SJSU library), vendor architecture material and current industry reports, assigned per unit and posted to Canvas.

Recommended references:

- *Designing Data-Intensive Applications* — Martin Kleppmann
- *Cloud Computing: Concepts, Technology, Security, and Architecture*, 2nd Edition — Erl, Monroy
- *Building Microservices*, 2nd Edition — Sam Newman
- *Fundamentals of Software Architecture* — Richards, Ford

**TO CONFIRM:** availability of each title in the current O'Reilly catalog.

---

## Course Requirements and Assignments

All course material is subject to a final examination. There is an interim midterm worth 10%; the final exam covers all topics from the semester and is worth 25%.

Students form groups of 2–4 for homework and term-project assignments.

### Term Project (33%)

| Component               | Format            | Weight |
|-------------------------|-------------------|--------|
| Project Report          | Word              | 3%     |
| Project Presentation    | PowerPoint        | 3%     |
| Project Demonstration   | Live demo         | 12%    |
| Project Features & Code | GitHub submission | 15%    |

## Assignments and Grading Policy

- Exams are a combination of multiple choice and short answer, **closed book**.
- Grades will be assigned on a curve.
- No extra credit.

### Grade Weighting

| Component                 | Weight      |
|---------------------------|-------------|
| Term Project              | 33%         |
| Final Exam                | 25%         |
| Homework (7 assignments)  | 22%         |
| In-Class Assignments (14) | 10%         |
| Midterm Exam              | 10%         |
| <b>Total</b>              | <b>100%</b> |

**Canvas configuration note (instructor — remove before distribution).** These are **group weights** and hold only if *Assignment Groups Weighted* is enabled in the Canvas gradebook, with each group set to the percentage above. In the prior offering every group weight was left at 0, so Canvas graded on raw points and the effective distribution was Project 28.9%, Homework 28.1%, Final 21.9%, In-Class 12.3%, Midterm 8.8% (114 raw points) — not what the syllabus stated.

### Grading Rules

| Grade | Percentage |
|-------|------------|
| A+    | 100%       |

| Grade | Percentage |
|-------|------------|
| A     | 93-99%     |
| A-    | 90-92%     |
| B+    | 87-89%     |
| B     | 83-86%     |
| B-    | 80-82%     |
| C+    | 77-79%     |
| C     | 73-76%     |
| C-    | 70-72%     |

## CMPE 272 — Fall 2026 Course Schedule

*Section 03 meets 15 times; section 49 meets 14 times and takes unit 14 asynchronously. The schedule is subject to change with fair notice; changes are announced via Canvas.*

| # | Sec 03 (Tue) | Sec 49 (Wed) | Topic                                                                                                                                                          | Notes / Deadlines |
|---|--------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 1 | Aug 25       | Aug 19       | <b>Introduction and Overview.</b> The enterprise stack end to end; how the pieces fit and where the money and the risk actually sit.                           |                   |
| 2 | Sep 1        | Aug 26       | <b>Enterprise Infrastructure.</b> Operating environments, enterprise networking, and the shift from server administration to managed platforms and containers. | HW1               |
| 3 | Sep 8        | Sep 2        | <b>Application Frameworks and APIs.</b> Enterprise                                                                                                             | HW2               |

| # | Sec 03 (Tue) | Sec 49 (Wed) | Topic                                                                                                                                                                                 | Notes / Deadlines |
|---|--------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
|   |              |              | application frameworks; API design, gateways and integration as a first-class architectural concern.                                                                                  |                   |
| 4 | Sep 15       | Sep 9        | <b>Cloud-Native Services.</b><br>Containers and orchestration; serverless and lambda services; service mesh — including an honest look at where mesh complexity is and is not repaid. |                   |
| 5 | Sep 22       | Sep 16       | <b>Platform Engineering and the Delivery Lifecycle.</b><br>Internal developer platforms; software development models and lifecycles; CI/CD and the platform-as-product shift.         | HW3               |
| 6 | Sep 29       | Sep 23       | <b>Enterprise Applications: ERP, SCM and CRM.</b> SAP, Salesforce and Dynamics; what these systems actually do, how they are                                                          | HW4               |

| #  | Sec 03 (Tue) | Sec 49 (Wed) | Topic                                                                                                                                                                    | Notes / Deadlines |
|----|--------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
|    |              |              | extended, and why replacing them is so hard.                                                                                                                             |                   |
| 7  | Oct 6        | Sep 30       | <b>Business Processes, Messaging and Service Composition.</b><br>Process modelling and BPMN; messaging and event-driven integration; composing services across systems.  |                   |
| 8  | Oct 13       | Oct 7        | <b>Project Abstract Presentations + Midterm Exam</b>                                                                                                                     | Midterm in class  |
| 9  | Oct 20       | Oct 14       | <b>Datastores, Databases and Query Languages.</b><br>Relational, document, key-value and graph stores; polyglot persistence; choosing a store against an access pattern. | HW5               |
| 10 | Oct 27       | Oct 21       | <b>Data Engineering and the Modern Data Stack.</b><br>Warehouses and lakehouses; ELT and transformation; batch versus streaming; data                                    | HW6               |

| #  | Sec 03 (Tue) | Sec 49 (Wed) | Topic                                                                                                                                                                                                                | Notes / Deadlines |
|----|--------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 11 | Nov 3        | Oct 28       | <p>quality and contracts.</p> <p><b>Enterprise AI I — Platforms and MLOps.</b></p> <p>Enterprise ML tooling and platforms; feature and model management; deployment, monitoring and drift.</p>                       | HW7               |
| 12 | Nov 10       | Nov 4        | <p><b>Enterprise AI II — LLM Integration.</b></p> <p>Retrieval-augmented generation over enterprise data; agents and tool use; evaluation, guardrails and governance; the cost of inference at enterprise scale.</p> |                   |
| 13 | Nov 17       | Nov 18       | <p><b>Enterprise Security, Identity and Supply Chain.</b></p> <p>Identity, authentication and authorization; zero trust; <b>software supply-chain integrity</b> — SBOM, signing and provenance.</p>                  | HW8               |

| #  | Sec 03 (Tue)                  | Sec 49 (Wed)                  | Topic                                                                                                                                                         | Notes / Deadlines        |
|----|-------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| 14 | Nov 24                        | <i>asynchronous</i>           | <b>Resilience, Observability and Monitoring.</b><br>Prometheus, ELK and Grafana; SLOs and error budgets; incident response and the operational feedback loop. |                          |
| 15 | Dec 1                         | Dec 2                         | <b>Final Project Presentations</b>                                                                                                                            | Last week of instruction |
| —  | <b>Tue, Dec 15:30-7:30 PM</b> | <b>Wed, Dec 95:30-7:30 PM</b> | <b>Final Exam</b> (SJSU Late Afternoon band, 5:30-6:25 PM start)                                                                                              | Rooms TBC                |

**Section 49 closures:** Nov 11 (Veteran’s Day) and Nov 25 (non-instructional day) — campus closed, no class. Unit 14 is posted to Canvas as an asynchronous module during this period.

---

## Classroom Protocol

- Classes start on time. If you miss a class, you are responsible for finding out what was announced and responding accordingly.
  - Please mute phones and avoid distracting activities during class.
  - Active participation in discussion is expected and is strongly associated with course grade.
  - Students may collaborate within their assigned group; work submitted must be that group’s own.
- 

## Use of Generative AI

- You must be able to explain any code or text you submit as if you had written it. You are accountable for it either way.
- Use of AI tools in homework and the term project must be disclosed, including the prompts and the model used.
- Tools may not replace individual effort or understanding of the material.

- Undisclosed use, or work you cannot explain, may be treated as academic dishonesty in accordance with University policy.

---

## University Policies

Per University Policy S16-9, university-wide policy information relevant to all courses, such as academic integrity, accommodations, etc., is available on the Office of Graduate and Undergraduate Programs' Syllabus Information web page at <http://www.sjsu.edu/gup/syllabusinfo/>

---

## Important Dates — Fall 2026

| Date               | Event                                                                        |
|--------------------|------------------------------------------------------------------------------|
| Wed, Aug 19        | First day of instruction                                                     |
| Mon, Sep 7         | <b>Labor Day — campus closed</b>                                             |
| Tue, Sep 15        | Last day to add / register late; last day to drop without an entry on record |
| Wed, Sep 16        | Enrollment census date                                                       |
| Wed, Nov 11        | <b>Veteran's Day — campus closed</b>                                         |
| Wed, Nov 25        | <b>Non-instructional day</b>                                                 |
| Thu–Fri, Nov 26–27 | <b>Thanksgiving / rescheduled holiday — campus closed</b>                    |
| Mon, Dec 7         | Last day of instruction                                                      |
| Tue, Dec 8         | Study/conference day — no classes or exams                                   |
| Dec 9–11, 14–15    | Final examinations                                                           |

*Source: SJSU Academic Year Calendar 2026/27.*